
CS 575: Design and Analysis of Algorithms

Syllabus – Spring 2026

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Instructor Information

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- Research interests: Brain Mapping, Biomedical Image Processing, Clinical Application of Medical Imaging, Neuroimaging, Blood flow imaging, Pattern Recognition, Computer Vision, Information Retrieval
- Office Hours: 2:00 PM – 3:00 PM on Tuesday and Thursday or by appointment
- Office hours zoom link: <https://binghamton.zoom.us/j/91740175822>

TA Information

- Yankun Zhang
 - Office: N00, Engineering Building
 - Email: yzhan401@binghamton.edu
 - Office Hours: 10am-12pm Wednesday or by appointment
 - Office hour zoom link: <https://binghamton.zoom.us/j/7477594084>

Learning Objectives

- Learn how to design and analyze algorithms!!!
 - Learn how to formulate a problem (Understand a real-world problem and translate it into a computational problem)
 - Learn how to choose an appropriate approach to design an efficient algorithm to solve the problem
 - Learn how to design an algorithm
 - Learn how to analyze algorithms and compare their time complexity
- ➔ *Build a solid foundation for your graduate study and career in CS by becoming very good at doing these things!*

Topics to be Covered in this Course

- Time Complexity Analysis and Growth Functions
- Divide and Conquer
- Dynamic Programming
- Greedy Algorithms
- Backtracking
- Branch and Bound
- Hashing
- Theory of NP
- Parallel Algorithms (if time permits)

Text Book & Lecture Notes

- Richard Neapolitan and Kumarss Naimipour, Foundations of Algorithms, 5th edition (ISBN: 978-1284049190)
- Cormen, Leiserson, Rivest and Stein, "Introduction to Algorithm", McGraw-Hill, 2009.
- S. Dasgupta, C.H. Papadimitriou, and U.V. Vazirani, "Algorithms", available at <http://highered.mheducation.com/sites/0073523402/index.html> "
- Lecture Notes
 - Lecture Notes (PowerPoint Slides) will be posted in the course folder on Brightspace

Prerequisite

- CS375 or equivalent (undergraduate algorithms and data structures)

Grading Policy

- Four Theoretical Homework Assignments: 20%
- Quizzes & Attendance: 5% (each)
- Midterms: 25%
- Final: 25%
- 4 Programming Assignments (implemented in C/C++/Java on `remote.cs.binghamton.edu`): 20%
- All late homework and projects will be penalized 10% per day (including weekends and holidays) and no assignment will be accepted, if it is more than 3 days late. Sometimes no late assignment will be accepted.
- Neither the TAs nor instructor will debug or review your code.
- If re-grading is requested, the entire assignment, project, or exam will be re-graded. As a result, your grade may decrease, increase, or stay the same.

Program Grading

- You will get a grade of at most 10% if your code does not compile. If the program compiles but it has runtime errors or bugs, your grade will be based on the severity of the bugs.
- Please make every effort to program and debug your code on your own. Use gdb, valgrind, IDE debuggers, and other tools. Use Google to learn about compiler and runtime error messages. Consult stackoverflow.com and other useful programming websites. As a last resort, please email questions to TA or visit office hours.

Basis of Grade Determination

The final grade for this course is largely based on your performance relative to the performance of other students in the class. As a reference, the approximate breakdown of grades from a previous class is summarized into the following table:

Grade	Overall Score
A	≥ 90
A-	$\geq 85 \text{ \& } < 90$
B+	$\geq 80 \text{ \& } < 85$
B	$\geq 75 \text{ \& } < 80$
B-	$\geq 70 \text{ \& } < 75$
C+	$\geq 67 \text{ \& } < 70$
C	$\geq 64 \text{ \& } < 67$
C-	$\geq 60 \text{ \& } < 64$
D	$\geq 55 \text{ \& } < 60$
F	< 55

Attendance

- Attendance of every class is required and will be checked on a random basis.
- An absence is counted if a student is not in the classroom when the attendance is checked.
- You may miss a total of three classes without a verifiable valid excuse.
- After that each unexcused absence will result in a penalty (5%) from the final overall score.
- If you have six or more unexcused absences, you automatically fail the course.
- Certain absences may be excused with prior consent from the instructor. Please inform the instructor of any expected absence in advance.

Expected Weekly Effort

- Classroom Instruction: 3 hours
- Outside classroom effort (reading, homework, and projects): approximately 6.5 hours

Academic Honesty

- Academic honesty and integrity are expected of every student. Dishonesty and cheating in all academic work related to this course, when discovered, will be severely punished.
- Read the Student Academic Honesty Code at <http://www2.binghamton.edu/watson/advising/pdfs/honesty-policy.pdf>
- Students must do their assignments/projects/exams by themselves. For the project report, students must write it using their own languages. All referenced works (including ideas, algorithms, programs, tables, figures, open source tools, etc.) must be clearly cited within the main body of the report and their full citations must be listed at the end of the report.
- Collaborations and general discussions are allowed and encouraged, but do not directly share your work with other students.

Academic Honesty (2)

- Each programming or theoretical assignment should start with the following statement:

“I have done this assignment completely on my own. I have not copied it, nor have I given my solution to anyone else. I understand that if I am involved in plagiarism or cheating I will have to sign an official form that I have cheated and that this form will be stored in my official university record. I also understand that I will receive a grade of 0 for the involved assignment for my first offense and that I will receive a grade of “F” for the course for any additional offense.”
- Each exam and quiz will have a first page with the following statement:

“I understand that if I am caught copying or talking during the exam/quiz I will have to sign an official form that I have cheated and that this form will be stored in my official university record. I also understand that I will receive a grade of 0 for the involved exam/quiz.”
- Your homework assignment, quiz or exam will not be graded unless the statement above is followed by your signature.

Classroom Etiquette

- No talking between students in class.
- Cell phones must be turned off or in vibrate mode.
- Laptop/notebook computers and tablets are only allowed for the purpose of taking notes during class.